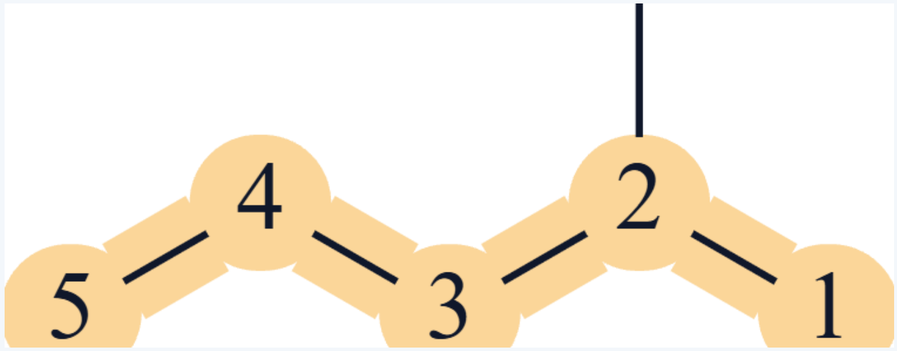
A01

FOUNDATION

Molecular formula (counted): **C6H14**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 5 C → pentane.

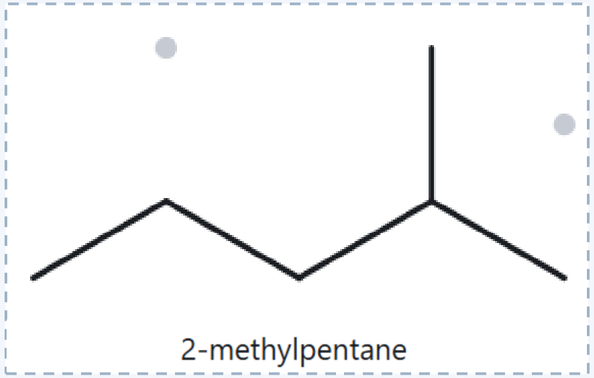
Numbering: right → left gives 2; left → right gives 4. First point of difference: 2 < 4, so number from the right.

Substituents: methyl on C2 → 2-methyl.

3 · MY NAME (WRITTEN FIRST)

2-methylpentane

4 · THE ALKANE CHECK



Returned name: 2-methylpentane

Verdict: ✓ **match** ☐ **mismatch**

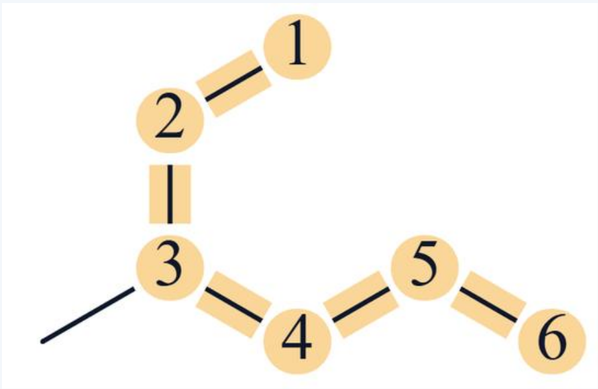
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A02

FOUNDATION

Molecular formula (counted): **C7H16**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 6 C → hexane (a bent chain, traced through the branch point).

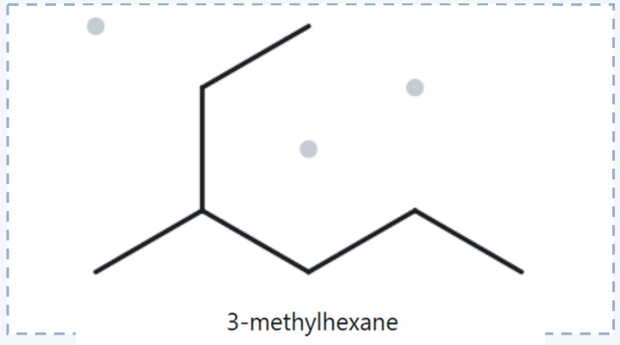
Numbering: from the top end gives 3; from the right end gives 4. First point of difference: 3 < 4, so number from the top.

Substituents: methyl on C3 → 3-methyl.

3 · MY NAME (WRITTEN FIRST)

3-methylhexane

4 · THE ALKANE CHECK



Returned name: 3-methylhexane

Verdict: ✓ **match** ☐ **mismatch**

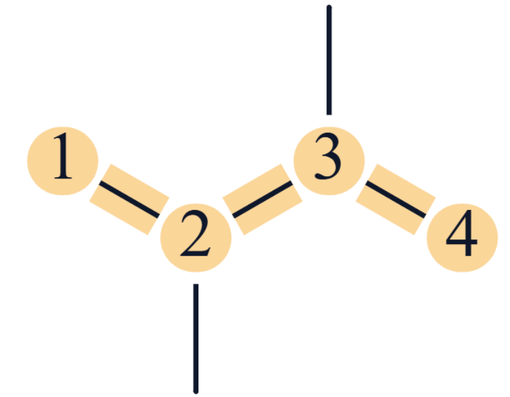
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A03

FOUNDATION

Molecular formula (counted): **C6H14**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 4 C → butane.

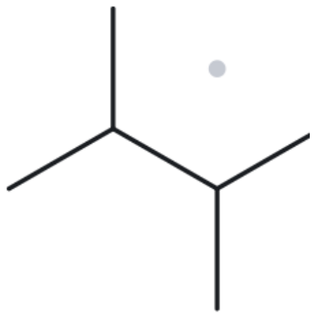
Numbering: both directions give 2,3, so either direction works.

Substituents (A→Z): two identical methyl groups on C2 and C3 → 2,3-dimethyl.

3 · MY NAME (WRITTEN FIRST)

2,3-dimethylbutane

4 · THE ALKANE CHECK



2,3-dimethylbutane

Returned name: 2,3-dimethylbutane

Verdict: ✓ **match** ☐ **mismatch**

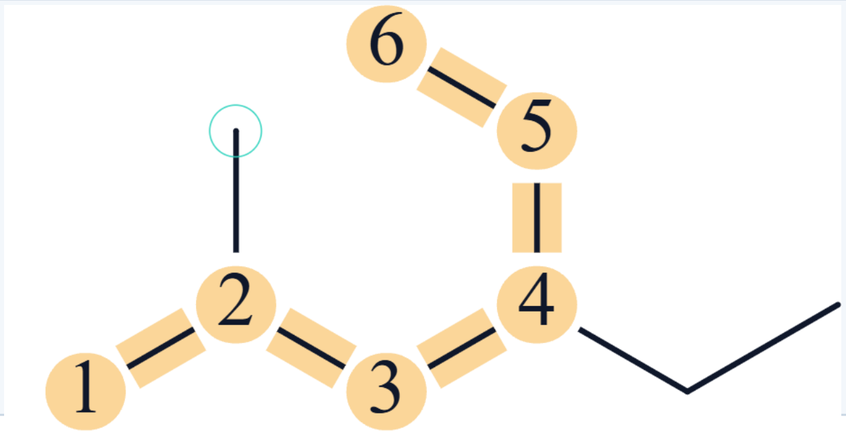
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A04

DEVELOPING

Molecular formula (counted): **C9H20**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 6 C → hexane. Two 6-C paths exist (through either two-carbon arm on C4), but both give the same name.

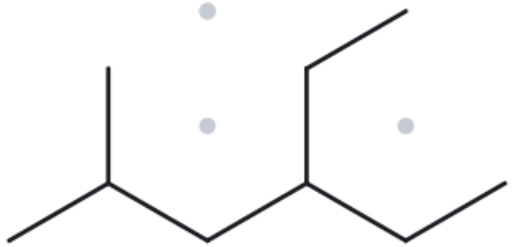
Numbering: lower left → upper right gives 2,4; the reverse gives 3,5. First point of difference: 2 < 3, so number from the lower left.

Substituents (A→Z): ethyl on C4, methyl on C2 → 4-ethyl-2-methyl (ethyl is cited first alphabetically).

3 · MY NAME (WRITTEN FIRST)

4-ethyl-2-methylhexane

4 · THE ALKANE CHECK



4-ethyl-2-methylhexane

Returned name: 4-ethyl-2-methylhexane

Verdict: ✓ **match** ☐ **mismatch**

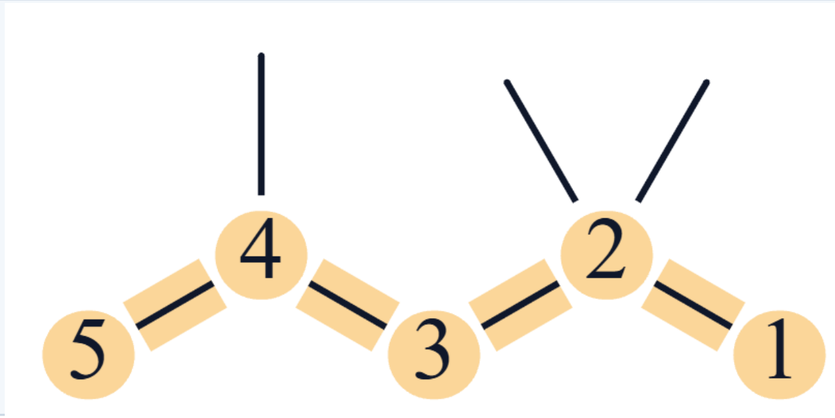
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A05

DEVELOPING

Molecular formula (counted): **C8H18**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 5 C → pentane.

Numbering: right → left gives 2,2,4; left → right gives 2,4,4. First point of difference: the second locant, 2 < 4, so number from the right.

Substituents: three methyl groups on C2, C2 and C4 → 2,2,4-trimethyl.

3 · MY NAME (WRITTEN FIRST)

2,2,4-trimethylpentane

4 · THE ALKANE CHECK



2,2,4-trimethylpentane

Returned name:

2,2,4-trimethylpentane

Verdict: ✓ **match** ☐ **mismatch**

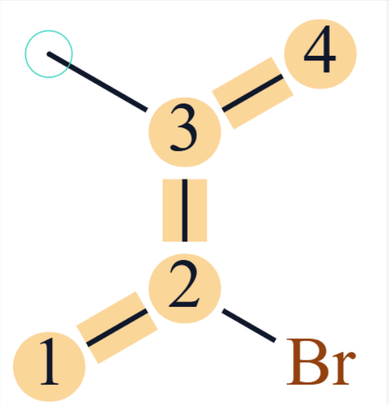
✓ **longest chain/ring** ✓ **lowest locants** ✓ **alphabetical order** ✓ **punctuation** ✓ **formula counted**

A06

DEVELOPING

Molecular formula (counted): **C5H11Br**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 4 C → butane.

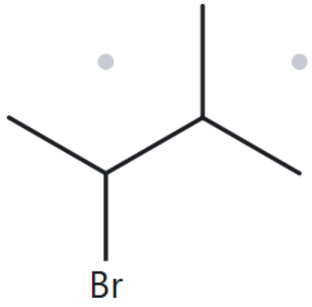
Numbering: bottom → top gives 2,3 (Br on C2, methyl on C3); top → bottom also gives 2,3. The locant sets tie, so the alphabetical tie-break applies: bromo comes before methyl, so Br gets the lower number.

Substituents (A→Z): bromo on C2, methyl on C3 → 2-bromo-3-methyl.

3 · MY NAME (WRITTEN FIRST)

2-bromo-3-methylbutane

4 · THE ALKANE CHECK



2-bromo-3-methylbutane

Returned name:

2-bromo-3-methylbutane

Verdict: ✓ **match** ☐ **mismatch**

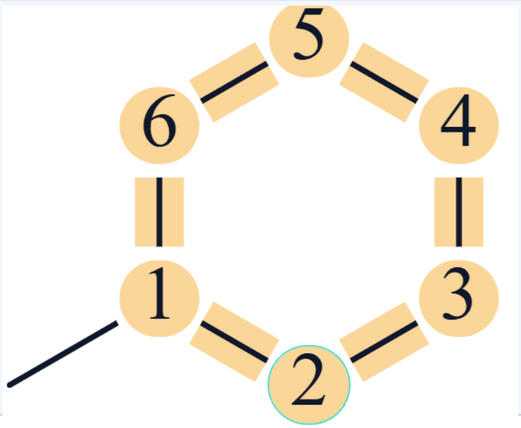
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A07

DEVELOPING

Molecular formula (counted): **C7H14**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: ring of 6 C → cyclohexane (the ring is the parent).

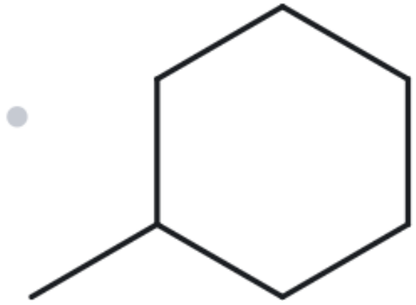
Numbering: with one substituent, the ring carbon bearing the methyl is C1 and the locant is not written.

Substituents: methyl → methylcyclohexane.

3 · MY NAME (WRITTEN FIRST)

methylcyclohexane

4 · THE ALKANE CHECK



methylcyclohexane

Returned name:

methylcyclohexane

Verdict: ✓ match ☐ mismatch

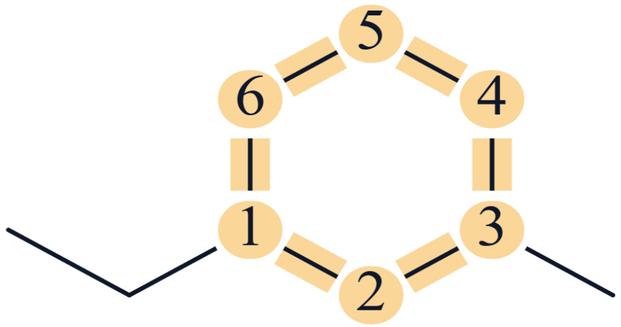
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A08

CHALLENGE

Molecular formula (counted): **C9H18**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: ring of 6 C → cyclohexane.

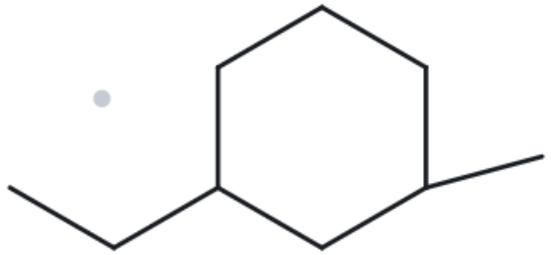
Numbering: either direction (the short way round) gives 1,3, so the locant sets tie. Alphabetical tie-break: ethyl comes before methyl, so ethyl is C1.

Substituents (A→Z): ethyl on C1, methyl on C3 → 1-ethyl-3-methyl.

3 · MY NAME (WRITTEN FIRST)

1-ethyl-3-methylcyclohexane

4 · THE ALKANE CHECK



1-ethyl-3-methylcyclohexane

Returned name:

1-ethyl-3-methylcyclohexane

Verdict: ✓ match ☐ mismatch

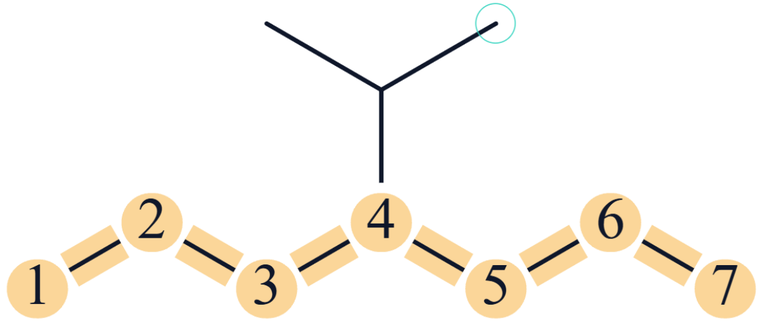
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A09

CHALLENGE

Molecular formula (counted): **C10H22**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 7 C, traced through the branch point (a path through the branch gives only 6 C) → heptane.

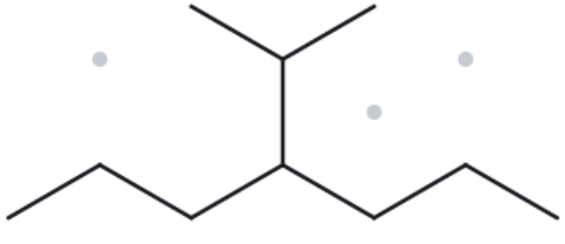
Numbering: either end gives 4, since the branch is on the central carbon.

Substituents: isopropyl, written (1-methylethyl), on C4 → 4-isopropyl.

3 · MY NAME (WRITTEN FIRST)

4-isopropylheptane

4 · THE ALKANE CHECK



2-methyl-3-propylhexane

Returned name:

2-methyl-3-propylhexane

Verdict: ☐ **match** ☒ **mismatch**

If ☒: correct name 4-isopropylheptane; rule that decides: longest chain (7 C, not the 6-C parent the tool gave). Same compound, different parent.

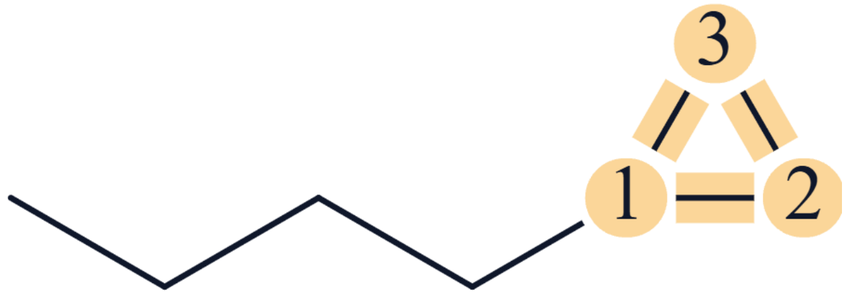
✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A10

CHALLENGE

Molecular formula (counted): **C7H14**

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: the ring is the parent in this activity (current IUPAC rule), even though the chain has 4 C and the ring 3 C → cyclopropane.

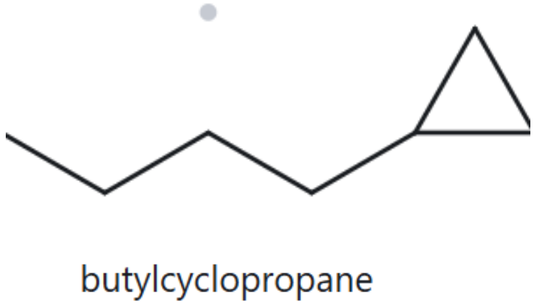
Numbering: C1 is the ring carbon bonded to the chain; with one substituent the locant is not written.

Substituents: butyl (the 4-carbon chain) → butylcyclopropane.

3 · MY NAME (WRITTEN FIRST)

butylcyclopropane

4 · THE ALKANE CHECK



Returned name:

butylcyclopropane

Verdict: ✓ **match** ☐ **mismatch**

✓ longest chain/ring ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

B01

FOUNDATION

Given name: **2-methylpropane**

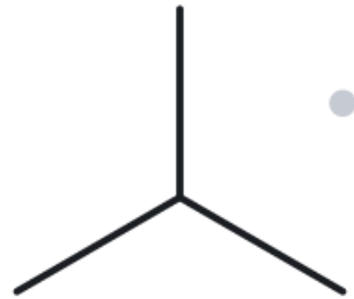
1 · NAME DECODED

Parent: Propane → Three (3) single-bonded C; chain

Substituents: methyl on C2

Expected formula: worked out from the name before drawing: C4H10

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C4H10

General formula: CnH2n+2 (open chain) = C4H10

Agrees? ✓ **yes** ☐ **no**

4 · BACK-CHECK

Returned name: 2-methylpropane

Verdict: ✓ **same as given** ☐ **differs**

✓ **parent length** ✓ **substituent positions** ✓ **formula agrees** ✓ **back-check recorded**

B02

FOUNDATION

Given name: **3-ethylpentane**

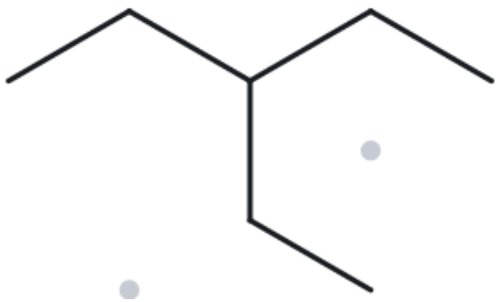
1 · NAME DECODED

Parent: Pentane → Five (5) single-bonded C; chain

Substituents: Ethyl on C3

Expected formula: worked out from the name before drawing: C7H16

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C7H16

General formula: CnH2n+2 (open chain) = C7H16

Agrees? ✓ yes ☐ no

4 · BACK-CHECK

Returned name: 3-ethylpentane

Verdict: ✓ same as given ☐ differs

✓ parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded

B03

FOUNDATION

Given name: **2,2-dimethylbutane**

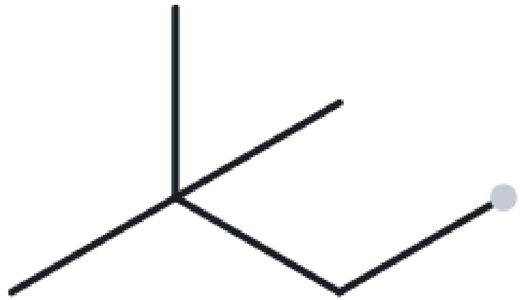
1 · NAME DECODED

Parent: Butane → Four (4) single-bonded C; chain

Substituents: methyl on C2 ; methyl on C2

Expected formula: worked out from the name before drawing: C6H14

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C6H14

General formula: CnH2n+2 (open chain) = C6H14

Agrees? ✓ yes ☐ no

4 · BACK-CHECK

Returned name: 2,2-dimethylbutane

Verdict: ✓ same as given ☐ differs

✓ parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded

B04

DEVELOPING

Given name: **3-ethyl-2-methylpentane**

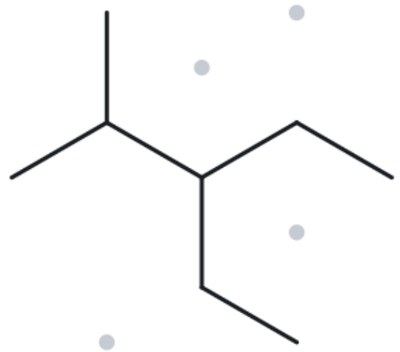
1 · NAME DECODED

Parent: Pentane → Five (5) single-bonded C; chain

Substituents: Ethyl on C3 ; methyl on C2

Expected formula: worked out from the name before drawing: C8H18

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C8H18

General formula: CnH2n+2 (open chain) = C8H18

Agrees? ✓ **yes** ☐ **no**

4 · BACK-CHECK

Returned name: 3-ethyl-2-methylpentane

Verdict: ✓ **same as given** ☐ **differs**

✓ **parent length** ✓ **substituent positions** ✓ **formula agrees** ✓ **back-check recorded**

B05

DEVELOPING

Given name: **1-chloro-2-methylpropane**

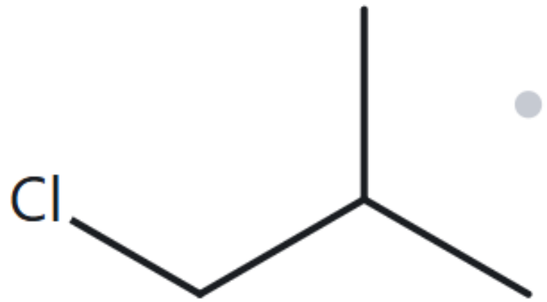
1 · NAME DECODED

Parent: Propane → Three (3) single-bonded C; chain

Substituents: Chloro on C1 ; methyl on C2

Expected formula: worked out from the name before drawing: C4H9Cl

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C4H9Cl

General formula: CnH2n+2 (open chain) = C4H10; each halogen replaces one H = C4H9Cl

Agrees? ✓ yes ☐ no

4 · BACK-CHECK

Returned name: 1-chloro-2-methylpropane

Verdict: ✓ same as given ☐ differs

✓ parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded

B06

DEVELOPING

Given name: **1,2-dimethylcyclopentane**

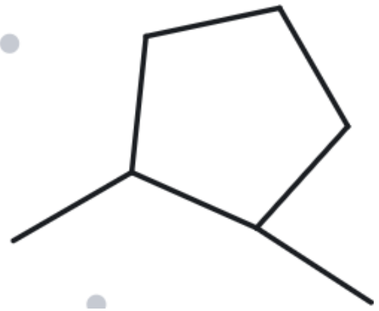
1 · NAME DECODED

Parent: Cyclopentane→ Five (5) single-bonded C; ring

Substituents: Methyl on C1 and C2

Expected formula: worked out from the name before drawing: C7H14

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C7H14

General formula: CnH2n (one ring) = C7H14

Agrees? ✓ **yes** ☐ **no**

4 · BACK-CHECK

Returned name: 1,2-dimethylcyclopentane

Verdict: ✓ **same as given** ☐ **differs**

✓ **parent length** ✓ **substituent positions** ✓ **formula agrees** ✓ **back-check recorded**

B07

DEVELOPING

Given name: **4-ethyl-3,3-dimethylheptane**

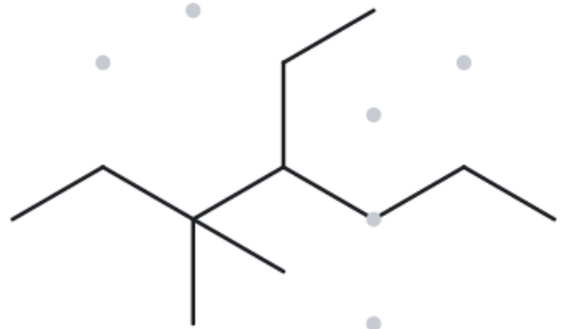
1 · NAME DECODED

Parent: Heptane → Seven (7) single-bonded C; chain

Substituents: Ethyl on C4 ; methyl on C3 (two)

Expected formula: worked out from the name before drawing: C11H24

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C11H24

General formula: CnH2n+2 (open chain) = C11H24

Agrees? ✓ yes ☐ no

4 · BACK-CHECK

Returned name: 4-ethyl-3,3-dimethylheptane

Verdict: ✓ same as given ☐ differs

✓ parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded

B08

CHALLENGE

Given name: **tert-butylcyclohexane**

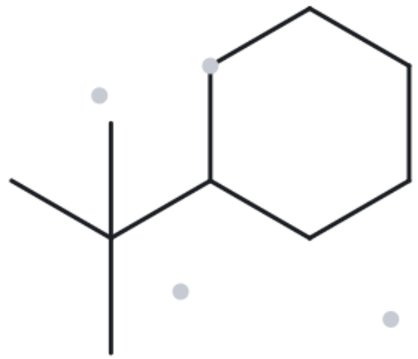
1 · NAME DECODED

Parent: Cyclohexane→ Six (6) single-bonded C; ring

Substituents: tert-butyl on C1

Expected formula: worked out from the name before drawing: C10H20

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C10H20

General formula: CnH2n (one ring) = C10H20

Agrees? ✓ **yes** ☐ **no**

4 · BACK-CHECK

Returned name: (1,1-dimethylethyl)cyclohexane

Verdict: ☐ **same as given** ✗ **differs**

Explanation: same compound, written with a different accepted substituent name. tert-butyl = (1,1-dimethylethyl) (Appendix B.3). No drawing error.

✓ **parent length** ✓ **substituent positions** ✓ **formula agrees** ✓ **back-check recorded**

B09

CHALLENGE

Given name: **3-ethyl-1,1-dimethylcyclohexane**

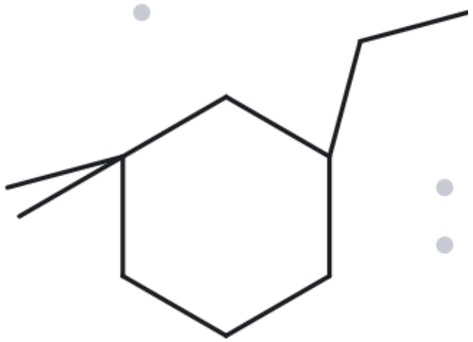
1 · NAME DECODED

Parent: Cyclohexane → Six (6) single-bonded C; ring

Substituents: Ethyl on C3 ; dimethyl on C1

Expected formula: worked out from the name before drawing: C10H20

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C10H20

General formula: CnH2n (one ring) = C10H20

Agrees? ✓ yes ☐ no

4 · BACK-CHECK

Returned name: 3-ethyl-1,1-dimethylcyclohexane

Verdict: ✓ same as given ☐ differs

✓ parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded

B10

CHALLENGE

Given name: **2-ethylbutane**

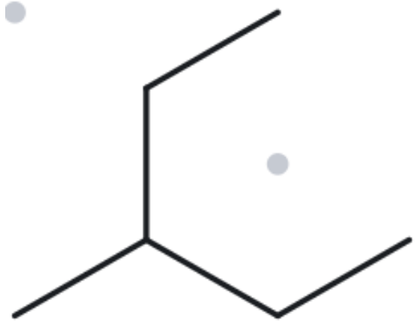
1 · NAME DECODED

Parent: Butane → Four (4) single-bonded C; chain

Substituents: Ethyl on C2

Expected formula: worked out from the name before drawing: C6H14

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from drawing: C6H14

General formula: CnH2n+2 (open chain) = C6H14

Agrees? ✓ yes ☐ no

4 · BACK-CHECK

Returned name: 3-methylpentane

Verdict: ☐ same as given ✗ differs

Explanation: This given name is deliberately incorrect.

Correct name: 3-methylpentane;

rule broken: the parent must be the longest continuous chain (the ethyl on C2 extends the chain to 5 C).

✓ parent length ✓ substituent positions ✓ formula agrees ✓ back-check recorded